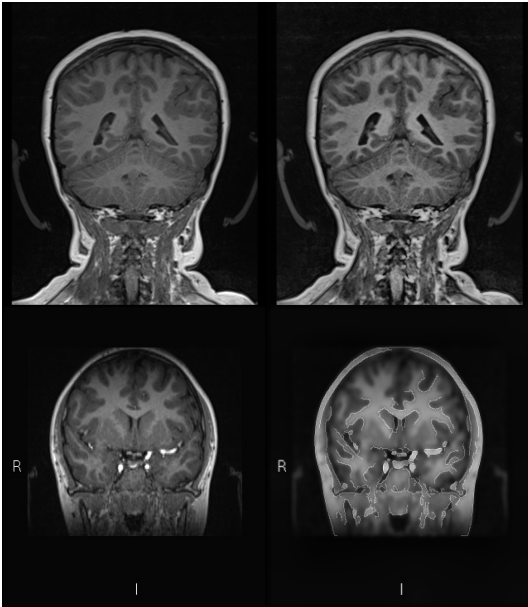




Specialcourse NRU

Holger Max Fløe Lyng

Investigating Methods for Polymicrogyria Classification

A replication of Guha et al. “Automated detection of polymicrogyria in pediatric patients using deep learning”

Dato: Juni 2026

Vejleder: Melania Ganz LLucia Coll Benejam

Dette er en skabelon for selve opstillingen. Kravene til hvilke oplysninger, der skal medtages, varierer fra fag til fag. Du kan derfor ikke uden videre benytte skabelonen til din egen afhandling. Du får nærmere oplysninger om kravene på dit eget studiekontor!

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1 [Purpose]

Article 2 - Preprocessing Reproduction (in progress). Reproduction of the preprocessing pipeline described in Article 2 is partially complete. Notably, the paper does not specify how differences in Field of View (FOV) or pixel/voxel spacing are handled across scans. As a first step, we will replicate their network architectures (ResNets and others) to verify whether we can reproduce the overfitting behavior reported in the paper.

2 Course motivation

3 Introduction

Deep learning has transformed the analysis of medical images, achieving performance comparable to clinical experts across different tasks including diabetic retinopathy grading, skin lesion classification, and radiological triage [?]tilføj litjens ref. In neuroimaging, convolutional neural networks have been applied to the automated detection of structural brain abnormalities from MRI, with the promise of supporting radiologists in high-volume screening and reducing diagnostic delays in remote areas. However, this requires that reported performances metrics are grounded in sound methodology, free from data leakage, label noise, and low-level image confounds that can inflate accuracy without reflecting genuine pathology signal detection.

Polymicrogyria(PMG) is a malformation of cortical development characterised by excessive cortical folding and abnormal lamination[6]. It is one of the most common malformations of cortical development, accounting for approximately 20% of all such malformations [7]. PMG is frequently encountered in patients with drug-resistant and is commonly associated with neurodevelopmental delay, motor impairments, and cognitive deficits. PMG remains challenging to characterize reliably due to its heterogeneous radiological appearance, variable distribution of anatomical features, and the limited availability of curated neuroimaging datasets.

Deep learning attempts to classify PMG have predominantly been conducted on small, imbalanced cohorts and without proper patient-level data splitting[4], making it difficult to distinguish genuine classification performance from artifacts of data leakage, class imbalance, or low-level image difference introduced during acquisition. Furthermore, the use of models pretrained on natural image datasets — comprising everyday photographs of cars, birds, and household objects — introduces a substantial domain gap relative to clinical neuroimaging; ImageNet-pretrained features have been shown to transfer poorly beyond the lowest convolutional layers, offering little benefit over task specific training [?]tilføj raghu ref om domæne gap, which may further limit the sensitivity of these models to the subtle cortical morphology features relevant to PMG classifica-

tion.

Automated detection of PMG from MRI could nonetheless inform clinical decision making by reducing the diagnostic burden on specialist neuroradiologists and enabling earlier identification of the condition. Fortunately, a publicly available benchmark dataset — the Pediatric Polymicrogyria MRI (PPMR) dataset [9] — has been released, and two independent studies have applied deep learning methods to PMG classification using this resource [9, 4].

This study undertakes an independent replication and methodological evaluation of guha et al. We examine the assumptions underlying their experimental design, assess whether the reported near perfect classification accuracy can be attributed to genuine PMG detection, and quantify the effect of applying corrective measure to the identified methodological issues. We identify three methodological errors in Guha et al.: (1) Label noise from including PMG-negative slices in the positive class, (2) a systematic class-correlated resolution difference that provides a shortcut signal, and (3) pre-split HC downsampling that produces artificially balanced evaluation sets and ignores patient-level integrity.

In addition, we evaluate the corrected pipeline on an independent clinical dataset of epilepsy patients, providing an assessment of generalization beyond the PPMR benchmark.

4 Background

- PMG background bases on <https://www.ncbi.nlm.nih.gov/books/NBK1329/andhttps://medlineplus.gov/genetics/condition/polymicrogyria/#resources>
- Maybe talk alittle about MRI
- Neural networks used. Resnet101 adn DenseNet201
- Maybe preprocessing steps or should that be en methods?

4.1 Polymicrogyria

Polymicrogyria (PMG) is a malformation of the developing brain characterized by abnormal cortical lamination and an excessive number of abnormally small gyri, resulting in an irregular folding pattern across all or part of the cerebral cortex [7][1]. PMG arises during the post-migrational stage of cortical development, distinguishing it from other malformations of cortical development (MCDs) such as lissencephaly, which affects the migration stage[6]. Multiple forms of PMG have been identified. Unilateral PMG affects one hemisphere only, whereas bilateral PMG involves both hemispheres, and severity ranges from mildly focal to extensive bilateral involvement. PMG accounts or approximately 20% of all MCDs, making it one of the most common brain malformations[7].

Despite its prevalence among MCDs, PMG remains challenging to diagnose reliably. MRI is the only imaging modality that provides sufficient soft-tissue contrast and spatial resolution to identify the fine cortical folds characteristic of PMG; computed tomography (CT) is inadequate for this task[6]. The radiological appearance of PMG is heterogeneous, e.g. features include an irregular or bumpy cortical surface, a stippled gray-white matter junction, and apparent cortical thickening. Some cases of PMG, adjacent microgyri fuse their overlying molecular layers, producing a deceptively smooth outer surface that can be mistaken for other MCDs. MRI appearance also changes with myelination, making diagnosis difficult in infants during the first year of life, when low gray-white matter contrast may obscure the slight cortical irregularities of the condition. Inter-rater variability among specialist neuroradiologists is well documented, underscoring both the difficulty of the diagnostic task and the potential value of automated detection methods.

4.2 MRI

Magnetic Resonance Imaging (MRI) is the principal structural neuroimaging modality across a wide range of neurological conditions, including epilepsy, brain tumors, white matter diseases, and cortical malformations. Its clinical dominance in neurology stems from its capacity to provide high soft-tissue contrast without ionising radiation. T1-weighted sequences in particular offer excellent differentiation between gray and white matter, enabling delineation of cortical thickness, sulcal morphology, and the gray-white matter junction architecture that is central to identifying MCDs such as PMG.

4.3 Deep learning for Medical Image Classification

4.3.1 Convolution Neural Networks

Convolution neural networks (CNNs) learn

4.3.2 ResNet-101

Residual learning was introduced by He et al. (2016)[?] to address the degradation problem in very deep networks, where accuracy saturates or degrades as depth increases. Each residual block learns a residual mapping $F(x)$ **add more...** so the block out is $H(x) = F(x) + x$. The skip connection allows gradients to flow directly through the network during backpropagation, making training of networks stable with over 100 layers. ResNet-101, comprising 101 weight layers organised in four residual stages achieved state-of-the-art performance on ImageNet ILSVRC 2015 and has become a widely adopted backbone for medical image and other classification tasks.

4.3.3 DenseNet-201

4.3.4 Transfer learning

4.4 Guha et al.

5 Methods

5.1 Data

5.1.1 PPMR

This project utilizes the publicly available PPMR dataset introduced by Zhang et al. [9] and subsequently used by Guha et al. [4]. The dataset consists of coronal 3D gradient-echo T1-weighted MRI slices from 23 pediatric epilepsy patients with confirmed PMG and three age and gender-matched healthy controls per patient, yielding a 3:1 control-to-case patient ratio.

A critical property of the dataset is that not all MRI slices from PMG patients contain visible malformation. Each slice was individually labeled by a pediatric neuroradiologist as PMG-positive (label 1), PMG-negative (label 2), or uncertain (label 3) [9]. As shown in Figure 1, only approximately 49% of all PMG-patient slices are labelled as PMG-positive; approximately 30% are PMG-negative, and approximately 19% are labeled uncertain.

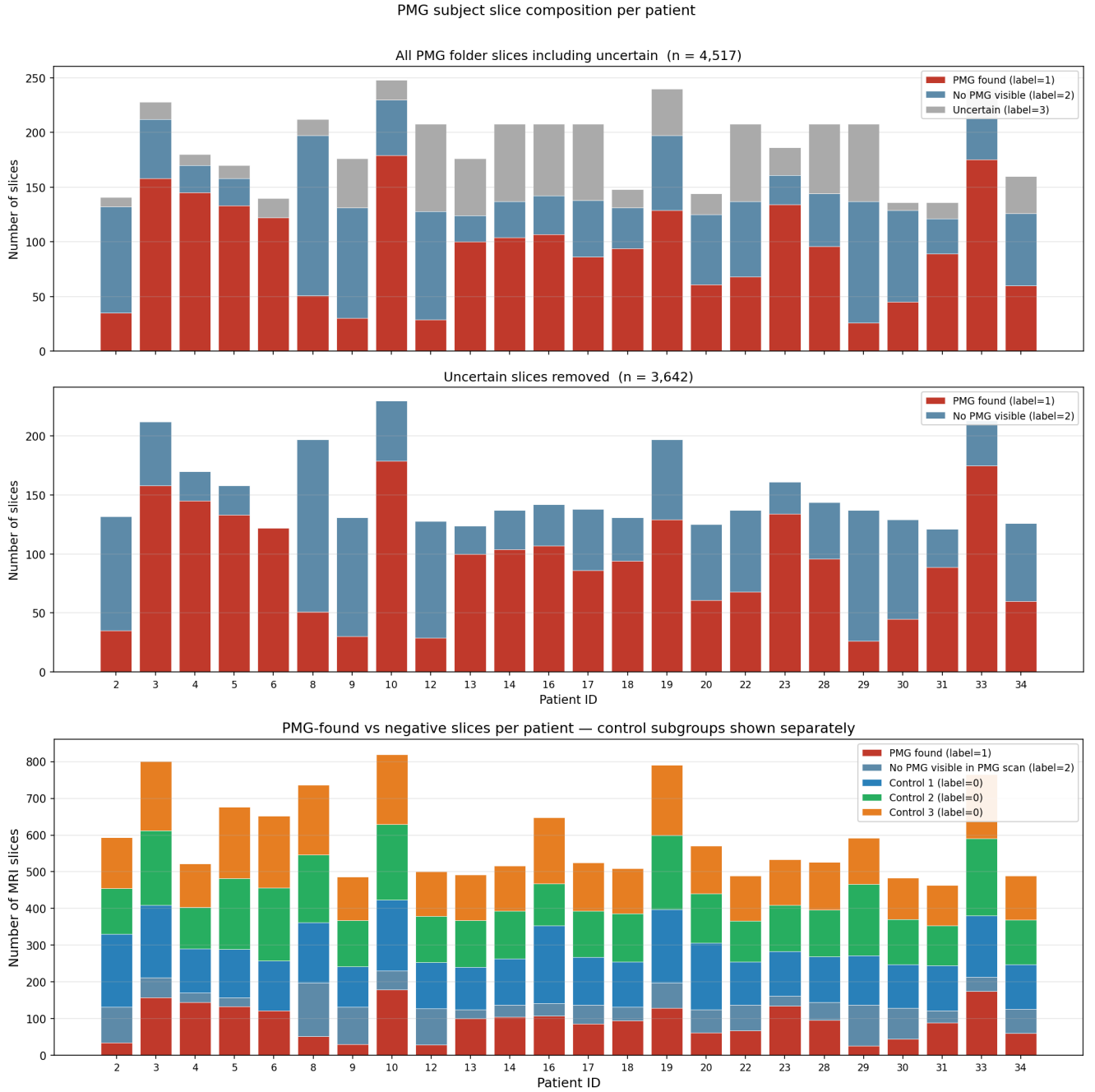


Figure 1: Top: Number of 2D slices per PMG patient labelled as PMG-positive (red), PMG-negative (blue), and uncertain (grey), based on the Kaggle release of the PPMR dataset [9]. Middle: The same distribution excluding uncertain slices, replicating the figure published in the original dataset paper [9]. Bottom: Total distribution of slices across control and PMG cohorts[9].

Aggregating across all labels, the dataset contains 4517 slices attributed to PMG patients and 10539 slices attributed to healthy controls. Guha et al. [4] report using precisely these counts in their analysis, treating all 4517 PMG-patient slices as a single positive class and all 10539 control

slices as a single negative class. The positive class contains a large proportion of non-pathology slices, which can dilute the PMG-specific attributes and introduce label noise into both training and evaluation. In figure 2 are the full statistics for the dataset.

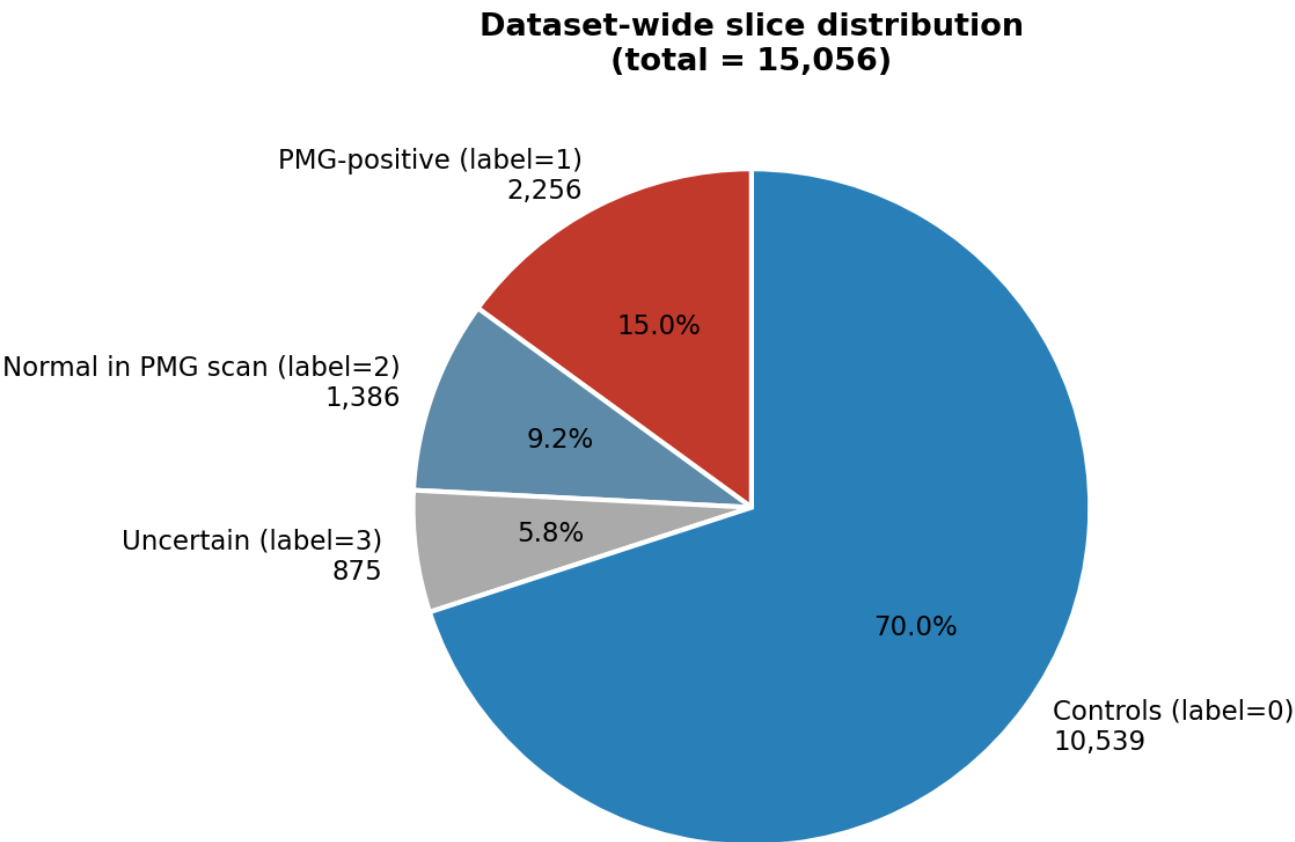


Figure 2: The full label distribution of the dataset published on Kaggle[9]. Guha et al. used this data distr

The dataset was labeled by a pediatric neuroradiologist with 12 of experience reading childrens brain MRIs. The patients was included in the study if PMG was found. The MRIs where a acquired using the highest resolution sequence for displaying cortical details, the coronal 3D gradient echo T1 weighted sequence, these volumes where exported as JPEG images from the PACS system. The imaging parameters used for acquisition of the coronal T1 weighted images are shown in Table

Parameter	3 T Siemens Skyra	1.5 T GE Cigna
Sequence	Coronal 3D GRE T1	Coronal 3D GRE T1
Repetition time (TR)	2200 ms	10.44 ms
Inversion time (TI)	1030 ms	450 ms
Echo time (TE)	2.63 ms	4.3 ms
Matrix	320 × 260	512 × 512
Slice thickness	1.2 mm	1.2 mm
Field of view (FOV)	20 × 23 cm	22 × 27 cm

Table 1: Acquisition parameters — PPMR dataset [9].

5.1.2 NRU data

To evaluate the preprocessing method and DL models used for the PMG classification task we introduce an in house dataset. The dataset includes 183 T1-weighted MRI scans from 162 unique patients evaluated for epilepsy (77 females; mean $\pm$ standard deviation (SD) age at acquisition: 25.8 ± 17.4 years). Of these, 110 scans (49 females; 24.7 ± 15.9 years) were reported to contain radiological evidence of PMG based on clinical radiology reports. The non-PMG group (37 females; 27.5 ± 19.6 years) consisted of scans in which PMG-related terminology appeared in the radiology report, but a subsequent double-rater confirmed the absence of PMG. Of these, 56 cases had other abnormalities and 17 were reported as having a normal MRI. The overall population distribution is shown in Fig. 1(c) **which fig? ask melanie**. MRI scans were acquired in routine clinical practice and retrospectively collected from the Eastern Region of Denmark, encompassing multiple scanner protocols, two magnetic field strengths (B), and three vendors. While no manual quality ratings were available, signal-to-noise ratio (SNR) was computed in white and gray matter as an objective measure of image quality. The mean $\pm$ SD SNR was 11.9 ± 2.7 in white matter and 7.9 ± 1.8 in gray matter, indicating that a variability in scan quality could contribute to errors in cortical surface reconstruction and feature extraction.

Parameter	Value
Scanner	Siemens Verio, 3 T
Sequence	MPRAGE (T1-weighted)
Repetition time (TR)	1900 ms
Echo time (TE)	2.23 ms
Inversion time (TI)	900 ms
Flip angle	9°
Voxel spacing	1 × 1 × 1 mm ³

Table 2: Acquisition parameters — Hvidovre Hospital dataset.

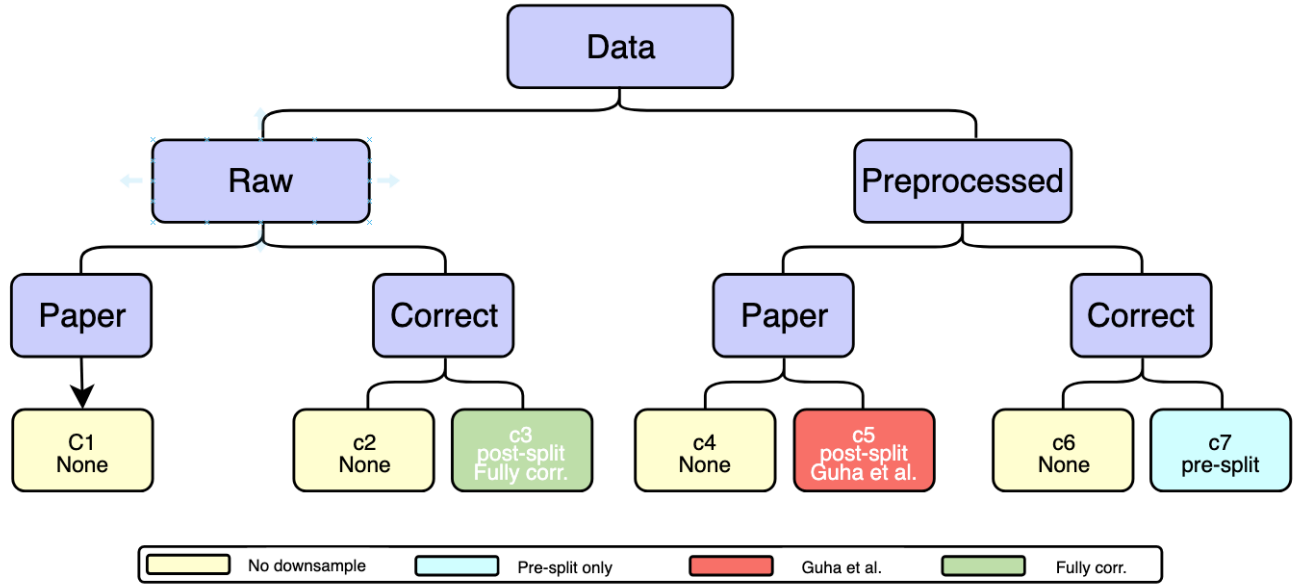


Figure 3: Caption

5.2 Experimental setup

To evaluate the methodological validity of Guha et al. (2025) and isolate the contribution of each identified error, experiments were organized into conditions spanning three axes: preprocessing (raw vs. preprocessed), label definition (paper vs. corrected), and downsampling strategy (none, pre-split, post-split). Both ResNet-101 and DenseNet-201 were evaluated under all conditions using 5-fold cross-validation at the patient level Table 5.2 provides an overview.

Condition	Preprocessing	Labels	Downsampling
1	Raw	Paper	None
2	Raw	Corrected	None
3	Raw	Corrected	Post-split
4	Preprocessed	Paper	None
5	Preprocessed	Paper	Pre-split
6	Preprocessed	Corrected	None
7	Preprocessed	Corrected	Pre-split

Table 3: Overview of the seven experimental conditions spanning three axes: preprocessing (raw vs. preprocessed), label definition (paper vs. corrected), and downsampling strategy (none, pre-split, post-split).

Following Guha et al. (2025), the papers label definition treats all 4517 slices attributed to

PMG patients as the positive class, regardless of their slice-level annotations, and all 10539 healthy control slices form the negative class. We updated the label definitions (corrected) restricting the positive class to slices with label = 1 ($n = 2256$). Label = 2 ($n = 1386$) are reclassified as negative examples and pooled with the original healthy controls, yielding $10539 + 1386 = 11925$ negative slices. Label = 3 slices are excluded from corrected conditions. All splits were performed at the patient level: the 92 unique subjects were partitioned such that all slices from a given patient appear exclusively in one split. A fixed random seed (seed = 42) ensured reproducibility.

The 10539 control slices were globally downsampled to 4517 before splitting, matching the class counts reported in their paper. The data was then split into training (60%), validation (20%), and test (20%) sets. We updated the label definition (corrected) to restrict the positive class to slices with label = 1 ($n = 2256$).

All splits in the conditions were performed at the patient level: the 92 unique subject were partitioned such that all slices from a given patient appear exclusively in one split. A fixed random seed (seed = 42) ensured reproducibility.

The experimental setup developed throughout the project. With further investigation into Guha et al. new experimental condition was developed. Overall seven condition were tested, structured to allow pairwise comparison of each error in isolation:

Then 5-fold cross validation at the patient level was used to obtain stable performance estimates for condition 3 and 4, to compare with Guha et al. results. Patients were divided by class and split into 5 folds, ensuring approximately equal class representation per fold. Within each fold, 15 % of the non-test patients were as a validation set; the models checkpoint with lowest validation loss was saved and evaluated on the test fold.

5.3 Preprocessing

The suggested preprocessing pipeline described in this section replicates, step-for-step, the procedure reported by Guha et al. [4] and was applied identically across all experiments condition and both datasets. The pipeline comprises five sequential operations applied to each 2D coronal T1-weighted MRI slices.

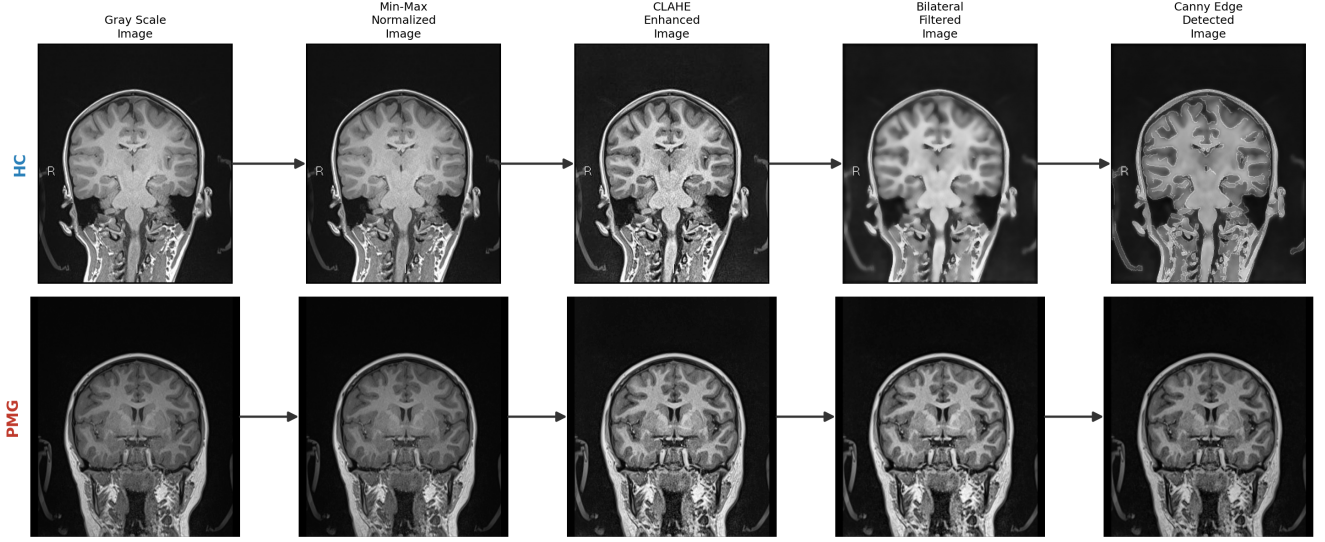


Figure 4: Caption

(1) Grayscale conversion. Each RGB JPEG image was converted to a single channel grayscale representation. This step reduces computational complexity in following operations and is a prerequisite for the grayscale specific filters that follow Guha et al.[4]. (2) Min-max normalization. Pixel intensities within each image were linearly rescaled to the interval $[0, 1]$ on a per image basis. This standardizes the range across scans acquired under heterogeneous imaging condition, ensuring that differences in scanner gain or acquisition protocol do not introduce systematic variation into the feature representation presented to the classifier [4]. (3) Contrast Limited Adaptive Histogram Equalisation (CLAHE). Local contrast was enhanced using CLAHE [10], with a clip limit of 2.0 and a grid size of 8×8 pixels. Unlike global histogram equalisation, CLAHE subdivides the image into non-overlapping contextual regions and applies histogram equalisation independently within each grid, enhancing local contrast variation that would otherwise be suppressed by global equalisation methods. The clip limit bounds the amplification applied to any histogram bin, preventing over amplification of noise characteristic of unconstrained adaptive methods[10]. For MRI of cortical architectures this localized enhancement should improve the visibility of fine structural details such as gyral folding patterns and gray-white matter boundaries [4] [5]. (4) Bilateral filtering. Noise reduction was achieved with bilateral filter [8], with a kernel size of 9×9 pixels and $\sigma_{color} = \sigma_{space} = 75$. The bilateral filter replaces each pixel's intensity with a weighted average of its spatial neighbourhood, where contributions are controlled both by a Gaussian spatial kernel and a Gaussian range kernel [8]. This dual-weighting mechanism suppresses high-frequency noise in homogeneous regions while preserving sharp transitions at anatomical boundaries – a property important for retaining cortical surfaces structure[2]. Following Guha et al [4], who

evaluated the bilateral filter against Non-Local Means, Anisotropic Diffusion, and Wavelet Denoising, the bilateral filter was selected as the most effective method for preserving the cortical boundaries and structure. (5) Canny edge detection. The Canny edge detector computes an edge map [3], with a low threshold 50, a high threshold of 200, and a Sobel aperture size of 3. The Canny algorithm applies a multi-stage pipeline comprising Gaussian smoothing, gradient magnitude estimation, non-maximum suppression and hysteresis-based edge tracking, yielding thin, well localized edge contours with reliable noise rejection [3]. For PMG detection, accurate delineation of subtle cortical irregularities, including irregular gyri and abnormally shallow sulci, depends on precise edge localisation. The Canny detector was selected over the Sobel operation, Scharr operator, and Laplacian of Gaussian filter after evaluation, all of which produced less reliable results on cortical MRI, according to Guha et al. [4]. The resulting binary edge map was blended with bilaterally filtered image with $\alpha = 0.20$, subtly reinforcing structural boundaries without discarding the underlying intensity information.

5.4 Models and Training

Two pretrained deep convolutional neural networks architectures were evaluated, following Guha et al. (2025): ResNet-101 (He et al., 2016) and DenseNet-201 (Huang et al., 2017), both initialized with ImageNet weights. The final fully connected layer of each model was replaced by a two layer classification head: a linear projection to 256 units, ReLU activation, dropout ($p=0.5$), and a final linear layer to a single logit, e.g. PMG vs control, following the head design of Guha et al. (2025). The backbone was frozen and only the classification head was trained, following Guha et al. (2025). Input images were resized to 224×224 pixels and replicated across three channels to match the ImageNet input format. Standard data augmentation was applied during training: random resized crop, 80-100% of the image area, resized to 224×224 pixels, and random horizontal flip.

Models were trained using the Adam optimizer with a learning rate of 5×10^{-4} , weight decay of 1×10^{-3} , and a batch size of 32, for 20 epochs, with no early stopping. Binary cross-entropy with logits (BCEWithLogitsLoss) was used as the loss function across all conditions. Performance was evaluated using accuracy, precision, recall, F1 score and Cohen's κ .

5.5 Ablation study

To probe whether trained models rely on spatially localised image features or on global low-level statistics, a black-box occlusion ablation was applied to all retained checkpoints. For each test image, a random square patch with a side length equal to 20% of the shorter image dimension was zeroed out prior to inference. The occluded test set was constructed once per split and then passed through every available checkpoint. Classification metrics were computed on the occluded

images and compared to the correspondings unoccluded test performance. A substantial drop in performance under occlusion is consistent with the model attending to specific spatial regions; a negligible drop suggest reliance on global texture or low-level statistics distributed across the image, which would further support the shortcut learning hypothesis.

6 Results

6.1 PPMR dataset

When investigating the PPMR dataset a confounding factor was almost immediately made clear, that the two cohort, control and PMG, had widely different resolution scale. This resolution confound is illustrated in figure ??, where it is easy to see a systematic difference in resolution scale. The control groups resolution ranges from 260×320 pixels to 512×512 pixels, where the latter is also being the median. The PMG groups resolution is approximately 1508×1727 pixels. The A second confounding factor is the systematic difference in image resolution between classes. As shown in Figure ??, PMG-patient images are exported from the PACS system at approximately 1508×1727 pixels, whereas healthy control images range from 260×320 pixels to 512×512 pixels. When all images are resized to the 224×224 pixel model input, PMG images are compressed by a factor of approximately $6\times$, while HC images are resized by only $1.1\text{--}2.3\times$. Heavy compression averages many original pixels into each output pixel, producing smooth, low-variance texture. Near-native resizing preserves the coarser, higher-variance texture of the original acquisition. A classifier can therefore distinguish the two classes simply by learning that smooth texture implies PMG and coarser texture implies HC – a signal that is present in every patch of every PMG-patient image, regardless of whether cortical pathology is visible in that region.

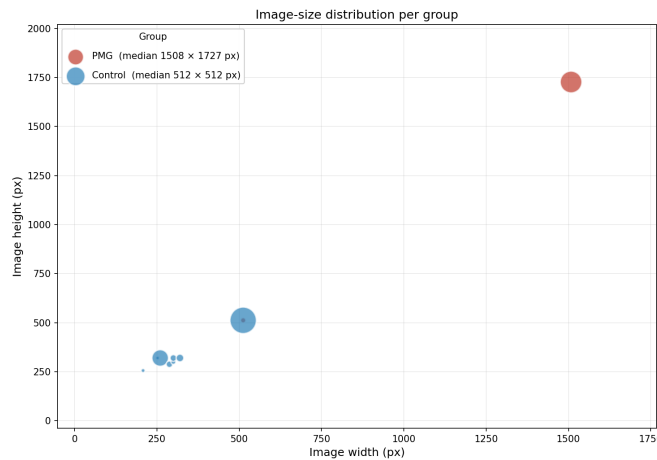


Figure 5: Caption

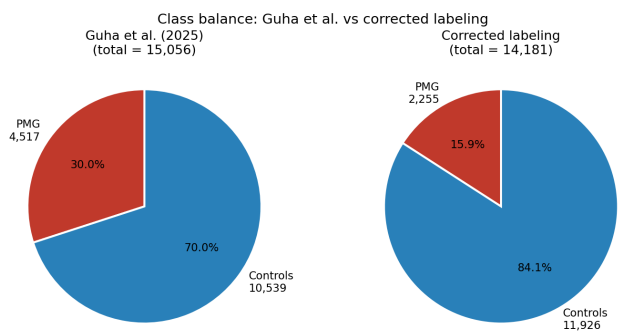


Figure 6: Caption

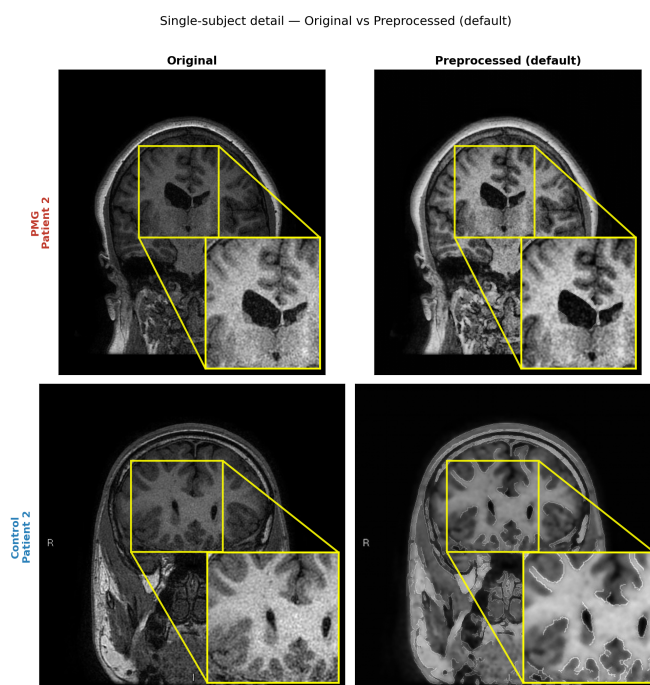


Figure 7: Caption

PMG group — Original vs Preprocessed (default)

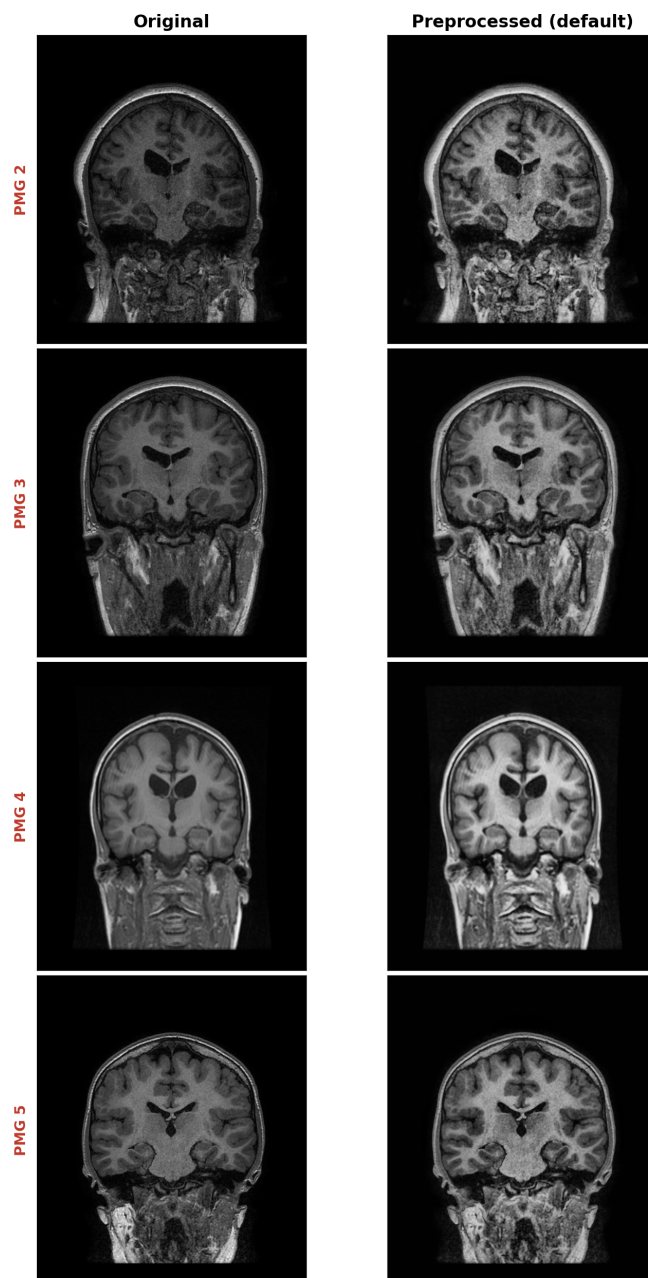


Figure 8: Caption

Control group — Original vs Preprocessed (default)

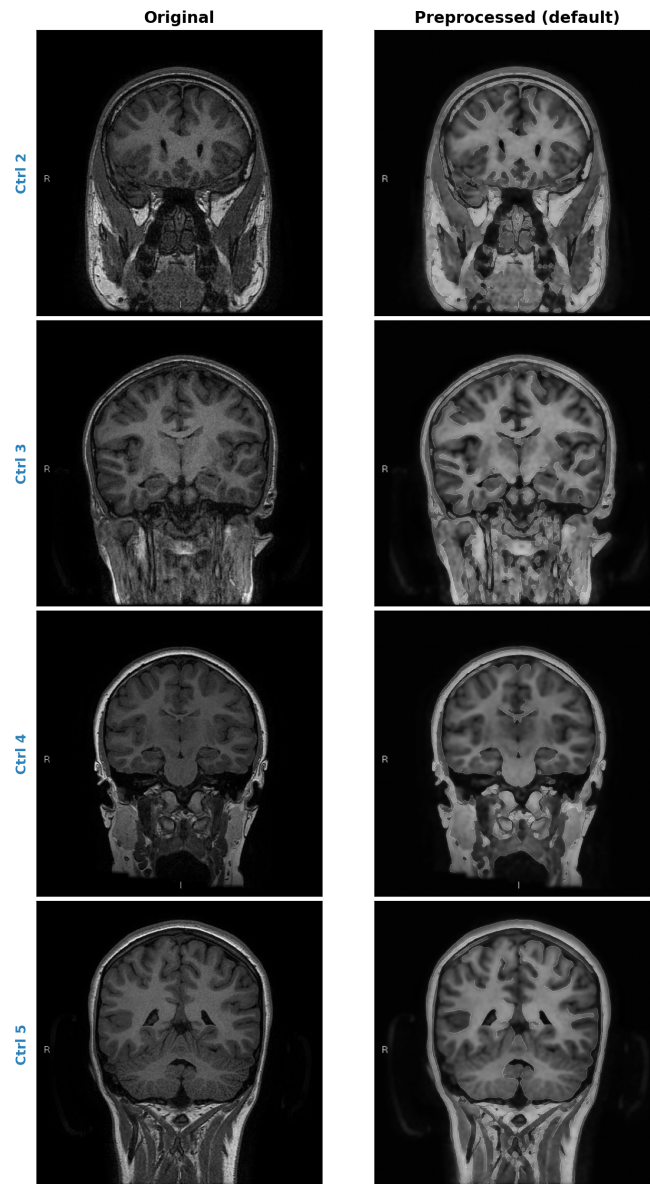


Figure 9: Caption

7 Discussion

8 Conclusion

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